ELEC 2FH4 - Bonus Assignment - Due March 8, 2026

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Set 1

Q3) Question 3,

Determine the total charge

(a) On line $0 < x < 5$ m if $\rho_L = 12x^2$ mC/m

(b) On the cylinder $\rho = 3, 0 < z < 4$ m if $\rho_S = \rho z^2$ nC/m^2

(c) Within the sphere $r = 4$ m if $\rho_v = \frac{10}{r \sin\theta}$ C/m^3

a)
$$Q = \int \rho_L \, dl$$
$$= \int_0^5 12x^2 \, dx$$
$$= \frac{12}{3} x^3 \Big|_0^5$$
$$= 4[5^3 - \cancel{0^3}] \text{ mC}$$
$$= 500 \text{ mC}$$
$$Q = 0.5 \text{ C}$$

b)
$$dS_{cylinder} = \rho \, d\phi \, dz, \quad \rho = 3$$
$$Q = \int \rho_s \, dS$$
$$= \int_0^4 \int_0^{2\pi} \rho z^2 \quad \rho \, d\phi \, dz$$
$$= \int_0^4 \int_0^{2\pi} \rho^2 z^2 \, d\phi \, dz$$
$$= \int_0^4 \int_0^{2\pi} 3^2 z^2 \, d\phi \, dz$$
$$= 9 \int_0^4 \int_0^{2\pi} z^2 \, d\phi \, dz$$
$$= 9 \, \frac{1}{3} z^3 \Big|_{\cancel{0}}^4 \, \phi \Big|_{\cancel{0}}^{2\pi}$$
$$= 9 \cdot \frac{1}{3}(4)^3 (2\pi)$$
$$= 384\pi \text{ nC}$$
$$= 1206 \text{ nC}$$
$$Q = 1.206 \, \mu\text{C}$$

c)
$$dV_{sphere} = r^2 \sin\theta \, dr \, d\theta \, d\phi$$
$$Q = \iiint \rho_v \, dV$$
$$= \int_0^4 \int_0^\pi \int_0^{2\pi} \frac{10}{\cancel{r \sin\theta}} \cdot r^{\cancel{2}} \cancel{\sin\theta} \, dr \, d\theta \, d\phi$$
$$= 10 \int_0^4 r \, dr \int_0^\pi d\theta \int_0^{2\pi} d\phi$$
$$= 10 \cdot \frac{1}{2} r^2 \Big|_{\cancel{0}}^4 \cdot \theta \Big|_{\cancel{0}}^\pi \cdot \phi \Big|_{\cancel{0}}^{2\pi}$$
$$= 10 \cdot \frac{1}{2}(4)^2 \cdot \pi \cdot 2\pi$$
$$= 160\pi^2$$
$$Q \doteq 1579 \text{ C}$$

Set 2

Q11) Question 11,

Determine the charge density due to each of the following electric flux densities:

(a) $\mathbf{D} = 8xy\mathbf{a}_x + 4x^2\mathbf{a}_y$ C/m^2

(b) $\mathbf{D} = 4\rho \sin\phi \, \mathbf{a}_\rho + 2\rho \cos\phi \, \mathbf{a}_\phi + 2z^2\mathbf{a}_z$ C/m^2

(c) $\mathbf{D} = \frac{2\cos\theta}{r^3}\mathbf{a}_r + \frac{\sin\theta}{r^3}\mathbf{a}_\theta$ C/m^2

$$\rho_v = \nabla \cdot D$$

a)
$$D = 8xy \, a_x + 4x^2 a_y \text{ C/m}^3$$
$$\rho_v = \nabla \cdot D$$
$$= \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$
$$= \frac{\partial}{\partial x} 8xy + \frac{\partial}{\partial y} 4x^2 + \frac{\partial}{\partial z} 0$$
$$= 8y + 0 + 0$$
$$\rho_v = 8y \text{ C/m}^3$$

b)
$$D = 4\rho \sin\phi \, a_\rho + 2\rho\cos\phi \, a_\phi + 2z^2 a_z \text{ C/m}^2$$
$$\rho = \nabla \cdot D$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho} \rho D_\rho + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho \cdot 4\rho \sin\phi) + \frac{1}{\rho}\frac{\partial}{\partial \phi}(2\rho\cos\phi) + \frac{\partial}{\partial z}(2z^2)$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}(4\rho^2 \sin\phi) + \frac{1}{\rho}\frac{\partial}{\partial \phi}(2\rho\cos\phi) + \frac{\partial}{\partial z}(2z^2)$$
$$= \frac{1}{\cancel{\rho}}(8\cancel{\rho}\sin\phi) + \frac{1}{\cancel{\rho}}(-2\cancel{\rho}\sin\phi) + 4z$$
$$= 8\sin\phi - 2\sin\phi + 4z$$
$$\rho_v = 6\sin\phi + 4z$$

c)
$$D = \frac{1}{r^3} 2\cos\theta \, a_r + \frac{1}{r^3}\sin\theta \, a_\theta \text{ C/m}^2$$
$$\rho_v = \nabla \cdot D$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta D_\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(D_\phi)$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^{\cancel{2}} \frac{1}{r^{\cancel{3}}} 2\cos\theta\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \cdot \frac{1}{r^3}\sin\theta\right) + \cancel{\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(0)}$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(\frac{1}{r} 2\cos\theta\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}\left(\frac{1}{r^3}\sin^2\theta\right)$$
$$= \frac{1}{r^2}\left(\frac{-1}{r^2} 2\cos\theta\right) + \frac{1}{r\cancel{\sin\theta}} \cdot \frac{1}{r^3} \cdot 2\cancel{\sin\theta}\cos\theta$$
$$= \frac{-1}{r^4} 2\cos\theta + \frac{1}{r^4} 2\cos\theta$$
$$\rho_v = 0$$

Set 3

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Q13) Question 13,

For each of the following potential distributions, find the electric field intensity and the volume charge distribution:

(a) $V = 2x^2 + 4y^2$
(b) $V = 10\rho^2 \sin\phi + 6\rho z$
(c) $V = 5r^2 \cos\theta \sin\phi$

- $\mathcal{E} = -\nabla V$
- $\rho_v = \nabla \cdot D = \nabla \cdot \varepsilon_0 \mathcal{E} \longrightarrow \rho_v = \varepsilon_0 \nabla \mathcal{E}$

a) $V = 2x^2 + 4y^2$

$$\mathcal{E} = -\nabla V$$
$$= -\left[\frac{\partial}{\partial x}(2x^2)a_x + \frac{\partial}{\partial y}(4y^2)a_y\right]$$
$$= -\left[4x\, a_x + 8y\, a_y\right]$$
$$\mathcal{E} = -4x\, a_x - 8y\, a_y$$

$$\rho_v = \varepsilon_0 \nabla \mathcal{E}$$
$$= \varepsilon_0\left[\frac{\partial}{\partial x}(-4x) + \frac{\partial}{\partial y}(-8y)\right]$$
$$= \varepsilon_0\left[-4 + -8\right]$$
$$= -12\varepsilon_0$$
$$\rho_v = -106.25\ pC/m^3$$

b) $V = 10\rho^2 \sin\phi + 6\rho z$

$$\mathcal{E} = -\nabla V$$
$$= -\left[\frac{\partial}{\partial \rho}(10\rho^2\sin\phi + 6\rho z)a_\rho + \frac{1}{\rho}\frac{\partial}{\partial \phi}(10\rho^2\sin\phi + 6\rho z)a_\phi + \frac{\partial}{\partial z}(10\rho^2\sin\phi + 6\rho z)a_z\right]$$
$$= -\left[(20\rho\sin\phi + 6z)a_\rho + \frac{1}{\cancel{\rho}}(10\rho^{\cancel{2}}\cos\phi)a_\phi + (6\rho)a_z\right]$$
$$\mathcal{E} = (-20\rho\sin\phi - 6z)a_\rho - 10\rho\cos\phi\, a_\phi - 6\rho\, a_z$$

$$\rho_v = \varepsilon_0 \nabla \mathcal{E}$$
$$= -\varepsilon_0 \nabla^2 V$$
$$= -\varepsilon_0\left[\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}\right]$$
$$= -\varepsilon_0\left[\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho(20\rho\sin\phi + 6z)\right) + \frac{1}{\rho^2}\frac{\partial^2}{\partial \phi^2}(10\rho^2\sin\phi + 6\rho z) + \frac{\partial^2}{\partial z^2}(10\rho^2\sin\phi + 6\rho z)\right]$$
$$= -\varepsilon_0\left[\frac{1}{\rho}\frac{\partial}{\partial \rho}(20\rho^2\sin\phi + 6\rho z) + \frac{1}{\rho^2}\frac{\partial}{\partial \phi}(10\rho^2\cos\phi + 0) + \frac{\partial}{\partial z}(0 + 6\rho)\right]$$
$$= -\varepsilon_0\left[\frac{1}{\rho}(40\rho\sin\phi + 6z) + \frac{1}{\rho^2}(-10\rho^2\sin\phi) + 0\right]$$
$$= -\varepsilon_0\left[40\sin\phi + \frac{1}{\rho}6z - 10\sin\phi\right]$$
$$\rho_v = -\varepsilon_0\left(30\sin\phi + \frac{1}{\rho}6z\right)\ C/m^3$$

c) $V = 5r^2\cos\theta\sin\phi$

$$\mathcal{E} = -\nabla V$$
$$= -\left[\frac{\partial}{\partial r}(5r^2\cos\theta\sin\phi)a_r + \frac{1}{r}\frac{\partial}{\partial \theta}(5r^2\cos\theta\sin\phi)a_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(5r^2\cos\theta\sin\phi)a_\phi\right]$$
$$= -\left[(10r\cos\theta\sin\phi)a_r + \frac{1}{\cancel{r}}(-5r^{\cancel{2}}\sin\theta\sin\phi)a_\theta + \frac{1}{\cancel{r}\sin\theta}(5r^{\cancel{2}}\cos\theta\cos\phi)a_\phi\right]$$
$$= -10r\cos\theta\sin\phi\, a_r - 5r\sin\theta\sin\phi\, a_\theta - 5\frac{\cos\theta}{\sin\theta}\cos\phi\, a_\phi$$
$$\mathcal{E} = -10r\cos\theta\sin\phi\, a_r - 5r\sin\theta\sin\phi\, a_\theta - 5\cot\theta\cos\phi\, a_\phi$$

$$\rho_v = -\varepsilon_0 \nabla^2 V$$
$$= -\varepsilon_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\cdot\frac{\partial^2 V}{\partial \phi^2}\right]$$
$$= -\varepsilon_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}(5r^2\cos\theta\sin\phi)\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial}{\partial \theta}(5r^2\cos\theta\sin\phi)\right) + \frac{1}{r^2\sin^2\theta}\cdot\frac{\partial^2}{\partial \phi^2}(5r^2\cos\theta\sin\phi)\right]$$
$$= -\varepsilon_0\left[\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\, 10r\cos\theta\sin\phi) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\, 5r^2(-\sin\theta)\sin\phi\right) + \frac{1}{r^2\sin^2\theta}\cdot\frac{\partial}{\partial \phi}(5r^2\cos\theta\cos\phi)\right]$$
$$= -\varepsilon_0\left[\frac{1}{r^2}\, 2r\, 10r\cos\theta\sin\phi) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}(-5r^2\sin^2\theta\sin\phi) + \frac{1}{r^2\sin^2\theta}\cdot\frac{\partial}{\partial \phi}(5r^2\cos\theta\cos\phi)\right]$$
$$= -\varepsilon_0\left[\frac{1}{\cancel{r^2}}\, 2\cancel{r}\, 10\cancel{r}\cos\theta\sin\phi) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}(-5r^2\sin^2\theta\sin\phi) + \frac{1}{r^2\sin^2\theta}\cdot\frac{\partial}{\partial \phi}(5r^2\cos\theta\cos\phi)\right]$$
$$= -\varepsilon_0\left[20\cos\theta\sin\phi + \frac{1}{\cancel{r^2\sin\theta}}(-10\cancel{r^2}\cancel{\sin\theta}\cos\theta\sin\phi) + \frac{1}{\cancel{r^2}\sin^2\theta}(-5\cancel{r^2}\cos\theta\sin\phi)\right]$$
$$= -\varepsilon_0\left[20\cos\theta\sin\phi - 10\cos\theta\sin\phi - 5\csc^2\theta\cos\theta\sin\phi\right]$$
$$\rho_v = -\varepsilon_0\left[10\cos\theta\sin\phi - 5\csc^2\theta\cos\theta\sin\phi\right]\ C/m^3$$

Set 4

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Q15) Question 15,

A dipole has dipole moment $p = 2a_x + 6a_y - 4a_z$ $\mu C \cdot m$. If the dipole is located in free space at (2, 3, −1), find the potential at (4, 0, 1).

- $p = 2a_x + 6a_y - 4a_z$ $\mu C \cdot m$
- dipole @ $(2, 3, -1)$ in free space
- V @ $(4, 0, 1) = ?$

$$V = \frac{p \cdot (r - r')}{4\pi\varepsilon_0 |r - r'|^3}$$

$$r - r' = (4, 0, 1) - (2, 3, -1) = (2, -3, 2)$$

$$|r - r'| = \sqrt{2^2 + 3^2 + 2^2} = \sqrt{17}$$

$$p \cdot (r - r') = (2, 6, -4) \cdot (2, -3, 2) = 4 - 18 - 8 = -22$$

$$V = \frac{p \cdot (r - r')}{4\pi\varepsilon_0 |r - r'|^3} = \frac{-22\,\mu}{4\cancel{\pi} \frac{10^{-9}}{36\cancel{\pi}} \sqrt{17}^3} = \frac{-22\,\mu}{7.788 \times 10^{-9}} = -2.825 \times 10^{15}$$

$$V = -2.83 \text{ kV}$$

Set 5

Q16) Question 16,

If $V = 2x^2 + 6y^2$ V in free space, find the energy stored in a volume defined by $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, and $-1 \leq z \leq 1$.

- $V = 2x^2 + 6y^2$ V in free space
- $\mathcal{E}$ stored within:
  - ↳ $-1 \leq x \leq 1$
  - ↳ $-1 \leq y \leq 1$
  - ↳ $-1 \leq z \leq 1$

$$\mathcal{E} = -\nabla V = -\frac{\partial V}{\partial x} a_x - \frac{\partial V}{\partial y} a_y = -\frac{\partial}{\partial x}(2x^2) a_x - \frac{\partial}{\partial y}(6y^2) a_y = -4x\, a_x - 12y\, a_y \text{ V/m}$$

$$W = \frac{1}{2}\varepsilon_0\left(\int_v |\mathcal{E}|^2 \, dv\right)$$

$$= \frac{1}{2}\varepsilon_0\left(\int_v |-4x\, a_x - 12y\, a_y|^2 \, dx\,dy\,dz\right)$$

$$= \frac{1}{2}\varepsilon_0\left(\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} (16x^2 + 144y^2) \, dx\,dy\,dz\right)$$

$$= \frac{1}{2}\varepsilon_0\left[\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} (16x^2) \, dx\,dy\,dz + \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} 144y^2 \, dx\,dy\,dz\right.$$

$$= \frac{1}{2}\varepsilon_0\left[\frac{16}{3}x^3 \cdot y \cdot z \Big|_{-1}^{1}\Big|_{-1}^{1}\Big|_{-1}^{1} + \frac{144}{3} y^3 \Big|_{-1}^{1}\Big|_{-1}^{1}\Big|_{-1}^{1}\right)$$

$$= \frac{1}{2}\varepsilon_0\left[\frac{16}{3}((1)^3 - (-1)^3) \cdot (2) \cdot (2) + \frac{144}{3}((1)^3 - (-1)^3) \cdot (2) \cdot (2)\right]$$

$$= \frac{1}{2}\varepsilon_0\left[\frac{16}{3} 2 \cdot 2 \cdot 2 + \frac{144}{3} 2 \cdot 2 \cdot 2\right]$$

$$= \frac{1}{2}\varepsilon_0\left(\frac{1280}{3}\right)$$

$$= \frac{1}{2} \times \frac{10^{-9}}{36\pi} \times \frac{1280}{3}$$

$$= 1.8863 \times 10^{-9}$$

$$W = 1.89 \text{ nJ}$$

- ❖ ***Choose ONE question from each set. Clearly indicate the question number in your solution.***
- ❖ ***You must include your full name, student ID, and student number on every page of your submitted solutions.***

Set 1

Question 1,

Two identical uniform sheet charges with $\rho_s = 100\ \text{nC/m}^2$ are located in free space at $z = \pm 2.0$ cm. What force per unit area does each sheet exert on the other?

Question 2,

If $\mathbf{E} = 20e^{-5}y\,(\cos 5x\mathbf{a}_x - \sin 5x\mathbf{a}_y)$, find (*a*) $|\mathbf{E}|$ at $P(\pi/6, 0.1, 2)$; (*b*) a unit vector in the direction of $\mathbf{E}$ at P; (*c*) the equation of the direction line passing through P.

Question 3,

Determine the total charge

(a) On line $0 < x < 5$ m if $\rho_L = 12x^2$ mC/m

(b) On the cylinder $\rho = 3$, $0 < z < 4$ m if $\rho_S = \rho z^2\ \text{nC/m}^2$

(c) Within the sphere $r = 4$ m if $\rho_v = \dfrac{10}{r \sin\theta}\ \text{C/m}^3$

Question 4,

A cube is defined by $0 < x < \text{a}$, $0 < y < \text{a}$, and $0 < z < \text{a}$. If it is charged with $\rho_v = \dfrac{\rho_o x}{a}$, where ρ_o is a constant, calculate the total charge in the cube.

Question 5,

A uniform charge 12 μC/m is formed on a loop described by $x^2 + y^2 = 9$ on the $z = 0$ plane. Determine the force exerted on a 4 mC point charge at (0, 0, 4).

Question 6,

An annular disk of inner radius a and outer radius b is placed on the xy-plane and centered at the origin. If the disk carries uniform charge with density ρ_s, find E at (0, 0, h).

Question 7,

The gravitation force between two bodies of masses m_1 and m_2 is

$$\mathbf{F}_g = \frac{Gm_1m_2}{r^2}\mathbf{a}_r$$

where $G = 6.67 \times 10^{-11}$ N(m/kg)^2. Find the ratio of the electrostatic and gravitational forces between two electrons.

(Electron Mass: $m_e = 9.1 \times 10^{-31} kg$)

Set 2

Question 8,

Three point charges are located in the $z = 0$ plane: a charge $+Q$ at point $(-1, 0)$, a charge $+Q$ at point (1, 0), and a charge $-2Q$ at point (0, 1). Determine the electric flux density at (0, 0).

Question 9,

In free space, $\mathbf{E} = 12\rho z\cos\phi\,\mathbf{a}_\rho - 6\rho z\sin\phi\,\mathbf{a}_\phi + 6\rho^2\cos\phi\,\mathbf{a}_z$ V/m. Find the electric flux through surface $\phi = 90°$ $0<\rho<2$, $0<z<5$.

Question 10,

If $\mathbf{D} = \sin\theta\sin\phi\,\mathbf{a}_r + \cos\theta\sin\phi\,\mathbf{a}_\theta + \cos\phi\,\mathbf{a}_\phi$ nC/m^2, find: (a) the charge density at (2, 30°, 60°), (b) the flux through $r = 2$, $0 < \theta < 30°$, $0 < \phi < 60°$.

Question 11,

Determine the charge density due to each of the following electric flux densities:

(a) $\mathbf{D} = 8xy\mathbf{a}_x + 4x^2\mathbf{a}_y$ C/m^2

(b) $\mathbf{D} = 4\rho \sin\phi\, \mathbf{a}_\rho + 2\rho\cos\phi\,\mathbf{a}_\phi + 2z^2\mathbf{a}_z$ C/m^2

(c) $\mathbf{D} = \dfrac{2\cos\theta}{r^3}\mathbf{a}_r + \dfrac{\sin\theta}{r^3}\mathbf{a}_\theta$ C/m^2

Set 3

Question 12,

The potential distribution in free space is given by

$V = \rho^2 e^{-z}\sin\phi$ V

Calculate the electric field strength at $(4, \pi/4, -1)$.

Question 13,

For each of the following potential distributions, find the electric field intensity and the volume charge distribution:

(a) $V = 2x^2 + 4y^2$

(b) $V = 10\rho^2\sin\phi + 6\rho z$

(c) $V = 5r^2\cos\theta\sin\phi$

Set 4

Question 14,

Point charges Q and $-Q$ are located at $(0, d/2, 0)$ and $(0, -d/2, 0)$. Show that at point (r, θ, ϕ), where $r \gg d$,

$$V = \frac{Qd \sin\theta \sin\phi}{4\pi\varepsilon_o r^2}$$

Find the corresponding E field.

Question 15,

A dipole has dipole moment $p = 2\mathbf{a}_x + 6\mathbf{a}_y - 4\mathbf{a}_z$ μC · m. If the dipole is located in free space at (2, 3, –1), find the potential at (4, 0, 1).

Set 5

Question 16, 4.65

If $V = 2x^2 + 6y^2$ V in free space, find the energy stored in a volume defined by $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, and $-1 \leq z \leq 1$.

Question 17, 4.66

Find the energy stored in the hemispherical region $r \leq 2$ m, $\underline{0 < \theta < \pi}$, $\underline{0 < \phi < \pi}$ where

$$\mathbf{E} = 2r \sin\theta \cos\phi \, \mathbf{a}_r + r\cos\theta \cos\phi \, \mathbf{a}_\theta - r \sin\phi \, \mathbf{a}_\phi \text{ V/m}$$

exists.

Question 18, 4.68

In free space, $\mathbf{E} = y^2\mathbf{a}_x + 2xy\mathbf{a}_y - 4z\mathbf{a}_z$ V/m. Determine the energy stored in the region defined by $0 < x < 2$, $-1 < y < 1$, $0 < z < 4$.

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Q)

Bonus Question

Given the vector current density

$$\mathbf{J} = 10\rho^2 z\,\mathbf{a}_\rho - 4\rho\cos^2\phi\,\mathbf{a}_\phi \quad \mathrm{mA/m^2}$$

in cylindrical coordinates:

(a) Find the current density vector $\mathbf{J}$ at the point $\rho = 3,\ \phi = 30^\circ,\ z = 2$

(b) Determine the total current flowing outward through the cylindrical surface defined by

$$\rho = 3, \quad 0 < \phi < 2\pi, \quad 2 < z < 2.8.$$

• $\vec{J} = 10\rho^2 z\,\vec{a_\rho} - 4\rho\cos^2\phi\,\vec{a_\phi}$ mA / m²

a) at $\rho = 3$, $\phi = 30^\circ = \frac{1}{6}\pi$, $z = 2$, $\vec{J} = ?$

$\vec{J} = 10\rho^2 z\,\vec{a_\rho} - 4\rho\cos^2\phi\, a_\phi$

$= 10(3)^2(2)\,\vec{a_\rho} - 4(3)\cos^2(\pi/6)\, a_\phi$

$= 180\,\vec{a_\rho} - 4(3)\left(\frac{\sqrt{3}}{2}\right)^2 a_\phi$

$\vec{J} = 180\,\vec{a_\rho} - 9\,\vec{a_\phi}$ mA / m²

b) total outward current flow through surface: $\rho = 3$, $0 < \phi < 2\pi$, $2 < z < 2.8$?

• on cylindrical surface $\rho = 3$, the unit normal is $\vec{a_\rho}$ (outwards) : $dS = \rho\, d\phi\, dz\, \vec{a_\rho} = 3\, d\phi\, dz\, \vec{a_\rho}$

$I = \int_S J \cdot dS$

$= \iint J \cdot dS$

$= \int_2^{2.8}\int_0^{2\pi} 10\rho^2 z \cdot 3\, d\phi\, dz$

$= \int_2^{2.8}\int_0^{2\pi} 10(3^2) z \cdot 3\, d\phi\, dz$

$= \int_2^{2.8}\int_0^{2\pi} 270 z\, d\phi\, dz$

$= 270 \int_2^{2.8} z\, dz \int_0^{2\pi} d\phi$

$= 270 \cdot \frac{1}{2} z^2 \Big|_2^{2.8} \cdot \phi \Big|_0^{2\pi}$

$= 270 \cdot \frac{1}{\cancel{2}}(2.8^2 - 2^2) \cdot \cancel{2}\pi$

$= 1036.8\pi$

$\doteq 3257.2$ mA

$I \doteq 3.257$ A